

ACTIVITY 2

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Question 1 – Amounts (Bar Chart)

Scenario

A supermarket chain wants to compare the monthly sales revenue generated by different product categories during June 2026. The management wants to identify the best and worst performing categories using an appropriate visualization.

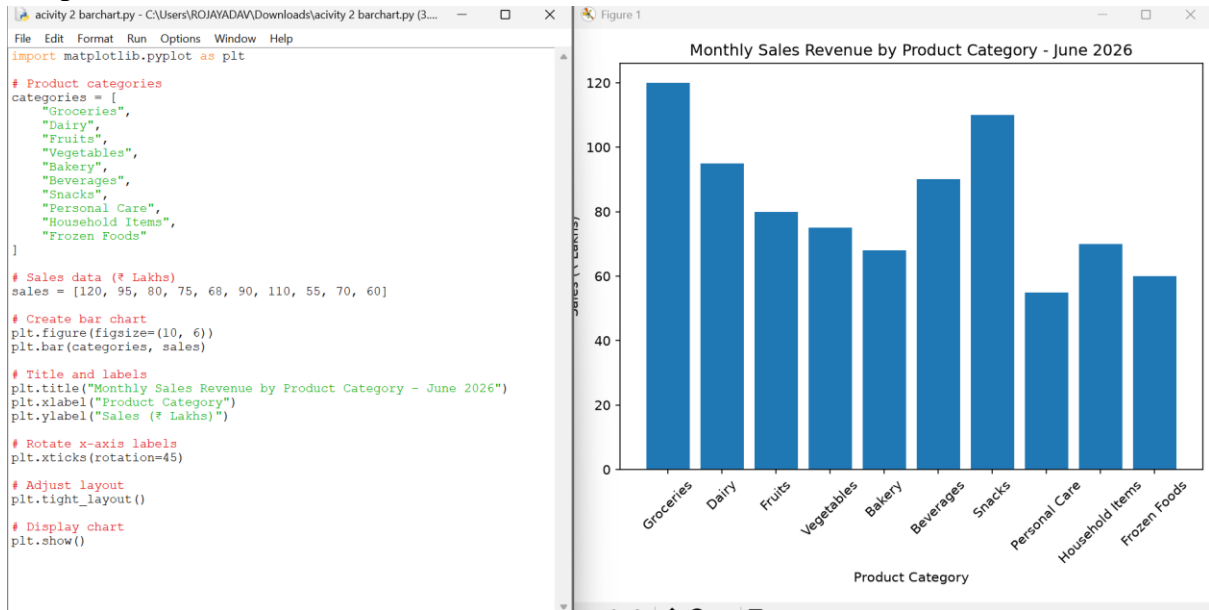
Dataset

Product Category	Sales (₹ Lakhs)
Groceries	120
Dairy	95
Fruits	80
Vegetables	75
Bakery	68
Beverages	90
Snacks	110
Personal Care	55
Household Items	70
Frozen Foods	60

Question

Create a **Bar Chart** to visualize the sales of each product category. Identify the highest and lowest selling categories and provide two observations based on the chart.

Output:



Question 2 – Distribution (Histogram)

Scenario

A university placement cell collected the annual salary packages (in LPA) offered to students during the campus recruitment drive. The placement officer wants to study the distribution of salary packages.

Dataset

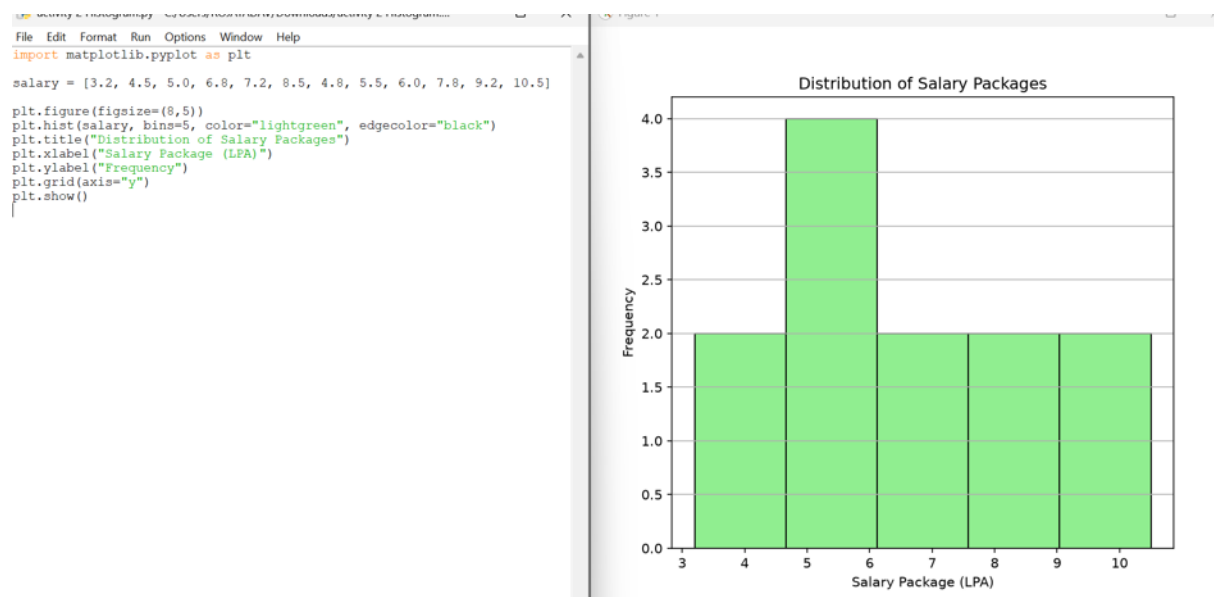
Student	Salary Package (LPA)
S1	3.2
S2	4.5
S3	5.0
S4	6.8
S5	7.2
S6	8.5
S7	4.8
S8	5.5
S9	6.0
S10	7.8

Student	Salary Package (LPA)
S11	9.2
S12	10.5

Question

Construct a **Histogram** using the above data. Analyze the distribution of salary packages and comment on the spread, frequency, and possible outliers

Output:



Question 3 – Proportions (Pie Chart)

Scenario

A software company allocated its annual digital marketing budget across different advertising channels. The marketing manager wants to visualize how the budget is distributed.

Dataset

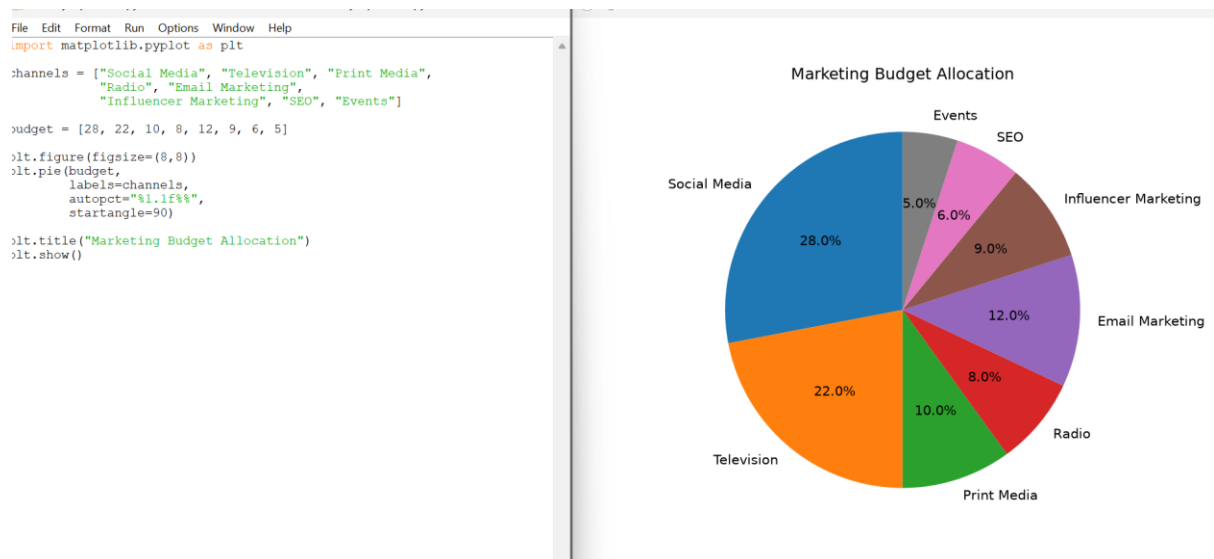
Marketing Channel	Budget (%)
Social Media	28
Television	22
Print Media	10
Radio	8
Email Marketing	12

Marketing Channel	Budget (%)
Influencer Marketing	9
SEO	6
Events	5

Question

Create a **Pie Chart** representing the marketing budget allocation. Identify the channel with the highest investment and explain the overall distribution.

Output:



Question 4 – X–Y Relationship (Scatter Plot)

Scenario

An e-commerce company wants to determine whether advertising expenditure influences monthly sales revenue.

Dataset

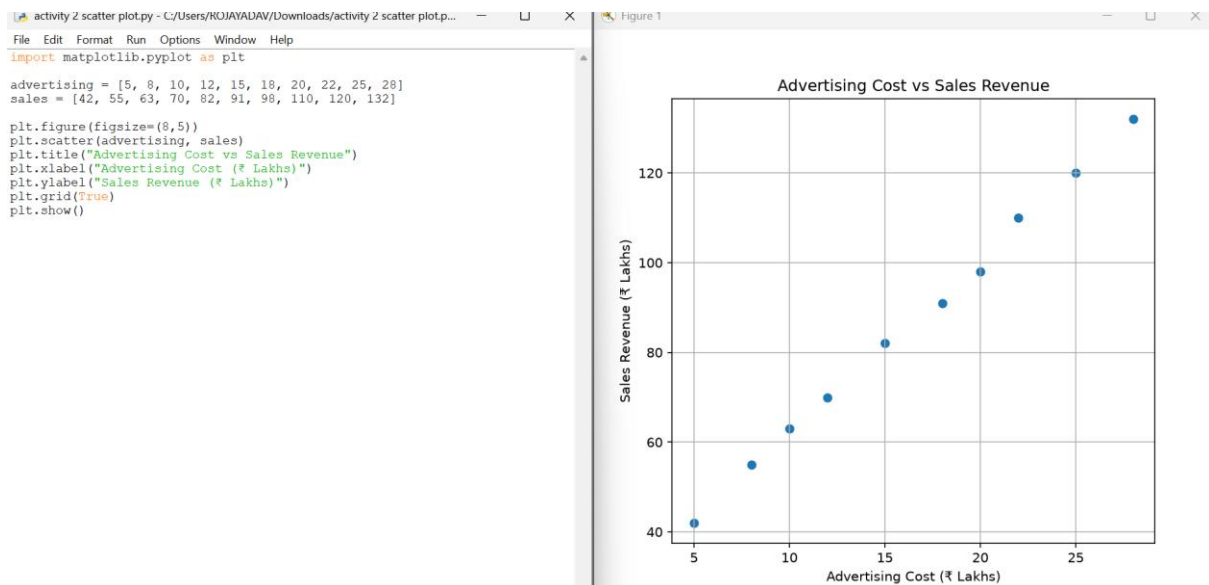
Month	Advertising Cost (₹ Lakhs)	Sales Revenue (₹ Lakhs)
Jan	5	42
Feb	8	55
Mar	10	63

Month	Advertising Cost (₹ Lakhs)	Sales Revenue (₹ Lakhs)
Apr	12	70
May	15	82
Jun	18	91
Jul	20	98
Aug	22	110
Sep	25	120
Oct	28	132

Question

Using the above dataset, create a **Scatter Plot** to examine the relationship between advertising expenditure and sales revenue. Determine whether the relationship is positive, negative, or weak, and justify your answer.

Output:



Question 5 – Amounts Over Time (Line Chart)

Scenario

A renewable energy company recorded the monthly electricity generated by its solar power plant throughout the year 2025. The company wants to study the trend in electricity generation.

Dataset

Month	Energy Generated (MWh)
January	420
February	450
March	490
April	530
May	560
June	600
July	580
August	590
September	610
October	630
November	590
December	540

Question

Construct a **Line Chart** using the above data. Analyze the monthly trend in electricity generation and identify the months with the highest and lowest energy production. Also comment on any seasonal variations observed in the data.

Output:

```
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt

months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun",
          "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]

energy = [420, 450, 490, 530, 560, 600, 580, 590, 610, 630, 590, 540]

plt.figure(figsize=(10,5))
plt.plot(months, energy, marker="o")
plt.title("Monthly Solar Energy Generation - 2025")
plt.xlabel("Month")
plt.ylabel("Energy Generated (MWh)")
plt.grid(True)
plt.show()
```

